Using Local VLMs for Data Extraction

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Rimu https://www.inaturalist.org/observations/198921399

Abstract

Global LLMs are readily available for research software engineering work, and researchers with a limited budget can easily use web interfaces to do complex tasks. However, most LLM providers are international corporations which can ignore or sidestep regional regulations or ethical concerns around sensitive data usage. For some RSEs, this blocks the usage of LLMs for development work. In this presentation, I will discuss why I used open source LLMs with open weights on a local grid provided by my institution to extract data from 65 years worth of PDFs from academic journals published in Aotearoa New Zealand. I focus on five main points that affect RSE work and researchers in Aotearoa New Zealand: Te Tiriti, funds, licensing, copyright, and reproducibility.

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Who is this Richard Littauer, anyway

- PhD Student at Te Herenga Waka
- Computer scientist
- Taxonomic nomenclaturist
- SustainOSS
- CURIOSS

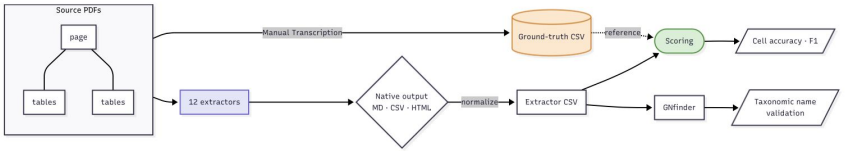


Fig. 3. The main PDF extractor pipeline.

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The Beach Patrol Scheme

BIRDS NEW ZEALAND
Te Kāhui Mātai Manu o Aotearoa

- Continuously running community-science survey
- Since 1943
- Annual summaries from 1960-2000, 2007-today
- 13k checklists; 100k rows; 30k records; 450k birds
- 207 taxa (including spuhs); 115~ seabirds
- Digitized between 1983-1990

https://www.birdsnz.org.nz/schemes/beachpatrolscheme/



Our goal

- Analyse BPS data for trends in beached birds
- Harmonize with eBird, iNaturalist, and GBIF.
- Predict future wrecks and HPAI influence.

My problem

The digitized database disagreed with public records.

So: I wrote code to redigitize 40 years of tables from PDFs.

Examples of Beach Patrol summaries

TABLE 1: Miles patrolled and birds found on different coasts

Coastline	2-Month Periods January to December J-F	M-A	M-J	J-A	S-O	N-D	Total Miles	Total Birds	Birds per Mile
Auckland East	10	8	4	1	7	4	34	38	1.1
Auckland West	35	17	6	94	15	–	167	478	2.9
Taranaki	–	–	14	5	12	4	35	9	0.3
Wgtn. West	25	18	50	66	42	74	275	483	1.8
Wgtn. South	3	2	20	9	3	5	42	61	1.5
Otago	–	–	10	6	12	15	43	19	0.4
Miscellaneous	–	7	8	3	7	17	42	33	0.8
Total Miles	73	52	112	184	98	119	638		
Total Birds	58	33	74	524	215	217		1121	
Birds per Mile	0.8	0.6	0.7	2.8	2.2	1.8			1.8

TABLE 2: List of Species Found

Species	Total Specimens	When Found (2-month periods) J-F	M-A	M-J	J-A	S-O	N-D	Where Found A	W
Eudyptula minor	61	2		4	20	18	17	32	28
E. albosignata	1				1			–	–
Eudyptes p. pachyrhynchus	1						1	–	1
Diomedea chrysostoma	4				3		1	3	1
D. bulleri	3			1	2			2	1
D. cauta	7		1	1		3	2	2	5
Diomedea sp.*	3		1		2			2	1
Phoebetria palpebrata	2				1	1		1	1
Macronectes giganteus	2					1	1	–	–
Daption capensis	11				7	3	1	8	2
Halobaena caerulea	2				1	1		1	1
Pachyptila vittata	16	4			8	3	1	6	9
P. salvini	34			1	32	1		29	5
P. desolata	18				16	2		13	5
P. belcheri	74				70	4		55	19
P. turtur	307	6		5	174	69	53	170	130
Pachyptila sp.*	52	5	1	3	14	15	14	9	42
Puffinus carneipes	4	2	1				1	3	–
P. bulleri	18	7	1	2	1	6	1	10	8
P. griseus	129	18	8	17	13	12	61	28	97
P. tenuirostris	8	2		4			2	2	6
P. g. gavia	145	1	7	2	87	43	5	98	41
P. g. huttoni	13						13	–	12
P. assimilis	2				1		1	1	1
Procellaria cinerea	1				1			1	–
P. westlandica	1				1			1	–
Pterodroma macroptera	2				1	1		1	1
P. lessoni	5		1			2	2	1	4
P. inexpectata	1		1					1	–
Pelecanoides urinatrix	38		1	6	15	10	6	8	28
Miscellaneous species†	156	11	10	28	53	20	34	28	95
TOTALS	1121	58	33	74	524	215	217	516	544

* Too fragmentary to allow specific identification.

† Other than petrels or penguins.

A—Auckland East & West Coasts.
W—Wellington West & South Coasts

Scope and solutions

172 separate tables needing transcription.

So:

- Use humans?
- Best OCR: ChatGPT-4 Vision (~90% accuracy)
- Other LLMs and VLMs

AIs used (forthcoming *eScience* paper)

TABLE I

COMPARATIVE EVALUATION RESULTS ON GT TABLES ($n = 21$). EACH METRIC CELL: MACRO MEAN [95% BCA CI] / CELL-WEIGHTED MEAN. PDFTOTEXT IS A RAW TEXT-DUMP CONTROL. FINAL COLUMN: CHANGE IN MACRO CELL-ACCURACY FROM 300→400-DPI RESCAN (PAIRED BCA); * = 95% CI EXCLUDES 0. 400-DPI CONTRASTS NOT ADJUSTED FOR MULTIPLICITY; TREAT * AS SUGGESTIVE. TEXT-LAYER TOOLS (CAMELOT, PDFPLUMBER, PDFTOTEXT) EXCLUDED; 400-DPI SCANS HAVE NO OCR LAYER.

Extractor	Cell accuracy	Row recall	Column recall	ΔCell acc. at 400dpi
Chandra (chandra-ocr-2)	**0.881** [0.78, 0.95] / 0.871	0.861 [0.71, 0.92] / 0.849	0.615 [0.45, 0.77] / 0.573	-0.098 [-0.25, +0.01]
Marker 1.0	0.535 [0.36, 0.69] / 0.544	0.829 [0.70, 0.91] / 0.818	0.562 [0.38, 0.72] / 0.584	-0.155 [-0.35, -0.00]*
olmOCR-2-7B	0.534 [0.36, 0.70] / 0.492	0.781 [0.55, 0.92] / 0.733	**0.721** [0.51, 0.86] / 0.673	+0.155 [+0.05, +0.32]*
Qwen3-VL-8B	0.492 [0.35, 0.63] / 0.461	0.704 [0.52, 0.83] / 0.625	0.712 [0.54, 0.84] / 0.701	+0.110 [-0.08, +0.30]
Qwen3.5-9B	0.346 [0.19, 0.53] / 0.324	**0.869** [0.76, 0.93] / 0.836	0.444 [0.25, 0.64] / 0.400	+0.176 [-0.03, +0.39]
Docling 2.0	0.276 [0.15, 0.46] / 0.219	0.561 [0.39, 0.70] / 0.507	0.405 [0.23, 0.60] / 0.386	+0.130 [-0.09, +0.31]
Qwen2.5-VL-7B	0.146 [0.05, 0.33] / 0.081	0.183 [0.05, 0.38] / 0.132	0.133 [0.04, 0.31] / 0.074	+0.136 [-0.03, +0.31]
Camelot 0.11	0.054 [0.00, 0.26] / 0.016	0.200 [0.09, 0.36] / 0.167	0.017 [0.00, 0.06] / 0.006	— *(text-layer)*
nougat-base	0.000 [0.00, 0.14] / 0.000	0.000 [0.00, 0.14] / 0.000	0.000 [0.00, 0.14] / 0.000	+0.000 [+0.00, +0.00]
pdfplumber 0.11	0.000 [0.00, 0.14] / 0.000	0.000 [0.00, 0.14] / 0.000	0.000 [0.00, 0.14] / 0.000	— *(text-layer)*
Pix2Struct-large	0.000 [0.00, 0.14] / 0.000	0.000 [0.00, 0.14] / 0.000	0.000 [0.00, 0.14] / 0.000	+0.000 [+0.00, +0.00]
Tesseract 5.3	0.000 [0.00, 0.14] / 0.000	0.383 [0.23, 0.54] / 0.394	0.000 [0.00, 0.14] / 0.000	+0.000 [+0.00, +0.00]
pdftotext (poppler) *(control)*	0.000 [0.00, 0.14] / 0.000	0.309 [0.16, 0.49] / 0.290	0.000 [0.00, 0.14] / 0.000	— *(text-layer)*

Toitū Te Tiriti

- I am debatably pākehā.
- What is the downstream usage of the data?
- Te Tiriti applies tino rangatiratanga to Māori, which includes taonga manu.
- Māori are not the only custodians of these species.
- I do not have collaborative ties on these issues.

These are ethical problems without solutions.

Funds

- The school grid is available to me.
- Using external credits sets up lock-in.
- Targeting tooling is cheaper.
- My work does not need to be expedited.

I am part of the funding ecosystem.

Licensing & Copyright

- Data disclaimer from BPS
- Transitive copyright issues
- *Notornis* is already online. But it also has a robots.txt.

Structured data favours existing power structures.

Reproducibility

- Online models are closed source or hidden behind a paywall
- Conversely: Open source models depend upon the ecosystem

CS is ½ reporting and ½ experimental methods.

- OSS is better for both.
- But: Claude makes me unsure of sharing my code.

Use it or lose it

- Using local LLMs forced upskilling
- Using local tools encourages funding their use.
- Demand leads to policy.

Qualms

- I used Claude to help me code the platform
- Claude significantly cut down on devops time
- I have no insight whether boundaries work (VSCode could lie)
- Tests?
- Collaboration was limited.
- *Notornis* is available online; does pointing matter?

Thanks.

Forthcoming: Richard Littauer, Kris Bubendorfer, Markus Luczak-Roesch. Data Rescue for Historic Scientific Literature: Automatically Extracting and Cross-Validating Published Records. *22nd IEEE International eScience Conference (eScience'26).*

Questions?



Kōkako https://www.inaturalist.org/observations/286751385